

Community Composting 101



Building a Compost Pile

Neighborhood
Soil Rebuilders
COMPOSTER TRAINING PROGRAM

ILSR
INSTITUTE FOR
Local Self-Reliance

Building a Compost Pile

- ✓ Recommendations for Materials to Accept & Avoid
- ✓ The Right Mix of Browns
- ✓ Measuring Moisture Content & Bulk Density
- ✓ Preparing Materials for Composting
- ✓ Methods for Building a Pile

A Guide to Best Management Practices



Table of Contents	
Introduction	6
PART 1: For New Composters and New Sites	
Composting Basics	16
Before You Begin	22
It Starts with a Plan	24
PART 2: The Science and Art of Composting	
Developing Your Composting Site	32
Building Your Composting Pile	34
PART 3: Managing the Composting Process and Your Site	
Turning or Mixing	41
Monitoring and Managing Pile	43
Finishing the Composting Process	47
Using Finished Compost in the Garden	51
Health and Safety Considerations	53
Community Composting Best Management Practices	55
Appendices	57
A. Glossary of Terms	57
B. Troubleshooting Pile	60
C. Troubleshooting Pile	64
D. Sample Composting Pile-Monitoring Plan	67
E. Community Composting in Georgia	69

<https://ilsr.org/composting-bmp-guide/>



Six Key Factors for Successful Composting

1. **MICROORGANISMS** that digest organic matter
2. The right balance of **FOOD**
 - Carbon or “Browns” for cell production
 - Nitrogen or “Greens” for energy
3. **WATER** to move & for biological functions
4. **OXYGEN** or air for aerobic respiration
5. The **SIZE** of the feedstock materials
6. The **VOLUME** of the pile

When all these needs are met, microbes will proliferate and produce heat.

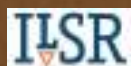


GREENS

Fresh materials
that are
relatively high
in nitrogen

BROWNS

Materials that
are relatively
high in carbon



ACCEPTABLE MATERIALS

✓ YES

GREENS



FRUIT & VEGETABLE SCRAPS
(No stickers)



EGG SHELLS



COFFEE GROUNDS & PAPER FILTERS



TEA BAGS
(No staples or plastic)



GARDEN TRIMMINGS
(6" or smaller)

BROWNS



FALL LEAVES



PLANT STALKS
(6" or smaller)



WOOD CHIPS & SHAVINGS
(Not chemically treated)



SHREDDED NEWSPAPER & BROWN BAGS
(No glossy pages)

✗ NO



MEAT, FISH, OR BONES



EGGS OR DAIRY PRODUCTS



PRODUCE STICKERS



GLOSSY PAPER



DISEASED AND PEST-INFESTED PLANTS



WEEDS WITH SEEDS



"COMPOSTABLE" TABLEWARE & PLASTIC BAGS



FATS, OILS, OR GREASE



COOKED FOOD



PET WASTE & KITTY LITTER



TREATED OR PAINTED WOOD



HERBICIDE-TREATED PLANTS



DRYER LINT



USED TISSUES

PFAS



Source: @LACompost

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GENERAL GUIDELINE

2-3 parts **BROWNS** to
1 part **GREENS** (by volume)



GREENS
(nitrogen-rich material)



BROWNS
(carbon-rich material)

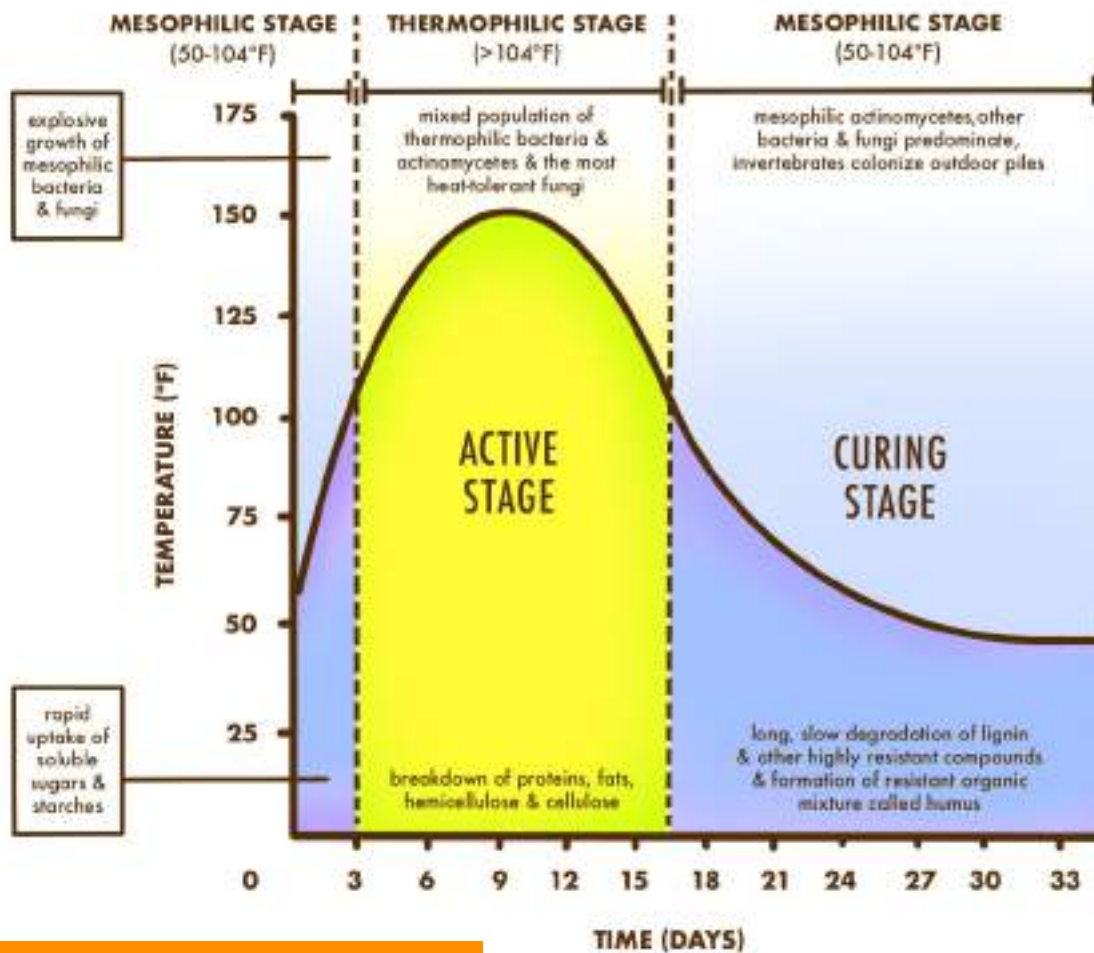
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The Right Mix of Browns

Earth Matter's browns recipe, courtesy of Earth Matter NY, Governors Island, New York City.
Source: ILSR



9



Different Phases, Different Microbes



The Right Mix of Browns

Earth Matter's browns recipe, courtesy of Earth Matter NY, Governors Island, New York City.
Source: ILSR



11

A SAMPLE RECIPE





Air & Bulk Density

Bulk density
= an estimation of porosity
= weight of a certain
volume of a material

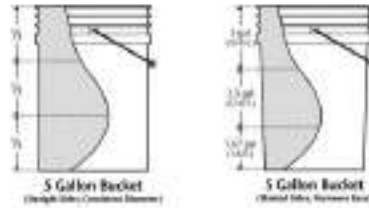
Starting
optimal range:
800 and 1,000
pounds
per cubic
yard



Calculating Compost Bulk Density

Materials Needed:

1. Compost pile
2. 5-gallon bucket (straight sides preferred)
3. Permanent pen marker
4. Gallon measuring container
5. Scale (pounds)
6. Ruler



Procedure:

1. With a 1-gallon measuring container, fill 5-gallon bucket with 5 gallons of water, as the trim of a "5 gallon" bucket is usually not 5 gallons. Mark this line on at least 3 places on the inside of the bucket with a permanent pen.
2. a) Buckets with straight sides (ie diameter at the top of the bucket is the same as the diameter at the bottom): measure height from bottom of inside of bucket to the 5 gallon line, divide this number by 3, and mark the bucket as above at 1 times and 2 times this number. For example, if the 5-gallon mark is 13 inches, dividing by 3 will get 4.33 inches, so mark that bucket at 4.33 inches and 8.67 inches.
b) If you cannot obtain a bucket with straight sides: find the two intermediate lines by measuring first 1 2/3 gallons (=1.67 gal or 6.31 liters or 1 gallon plus 10 2/3 cups) then 3 1/3 gallons (=3.33 gallons or 12.62 liters or 3 gallons plus 5 1/3 cups) of water into the bucket and marking where the water line is at both volumes with a pencil when wet, then a permanent pen when dry.
3. Compost used should be representative of the pile. Take handful samples (not large shovel full samples) from several locations in the pile. Dig into pile a couple of feet, do not take from dried-out outer layer of pile.
4. Fill bucket to the 1/3 line with compost. Drop bucket squarely from approximately 1 foot high to the ground (hard surface) 10 times.
5. Fill bucket to the 2/3 line. Drop bucket squarely from approximately 1 foot high to the ground 10 times.
6. Fill bucket to the 3/3 (5 gallon) line. Drop bucket squarely from approximately 1 foot high to the ground 10 times.
7. Fill bucket again to the 3/3 (5 gallon) line (do NOT drop bucket) and weigh in pounds.
8. Multiply weight of material in pounds (excluding bucket weight) by 40. This is your bulk density, in pounds per cubic yard.



Photo: ILSR

Washington State University:
<https://puyallup.wsu.edu/soils/bulk-density/>



THE "HAND SQUEEZE" MOISTURE TEST



Take handfuls from different places in your compost pile.

Give them a squeeze!

At the ideal moisture level, a few drops of water should appear between your knuckles like a wrung-out sponge.



Preparing Your Materials



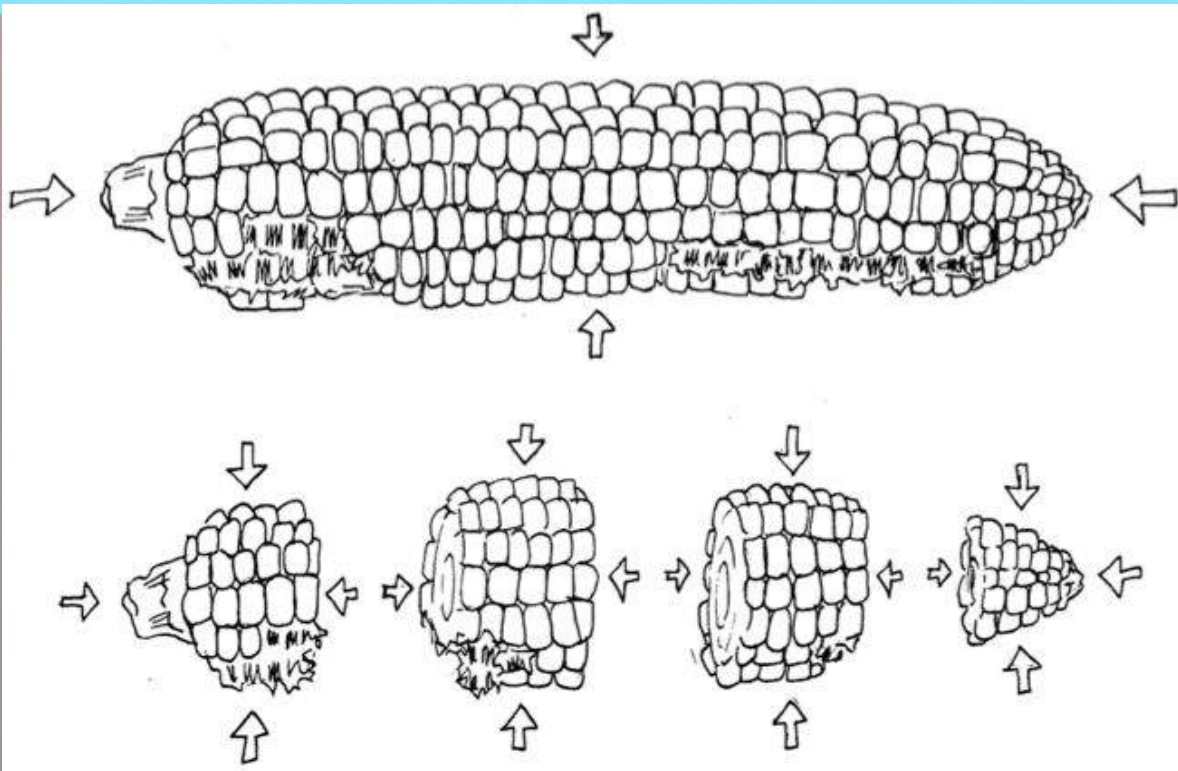
Participants practice chopping food scraps during an ILSR-led training at Northside Baptist Church in Baltimore.
Photo: ILSR

Size Matters!

- Chop as needed
- Increases surface area
- $\frac{1}{8}$ - 2 inches for fastest composting
- Aim for 2 to 6 inches for bigger materials



Preparing Your Materials



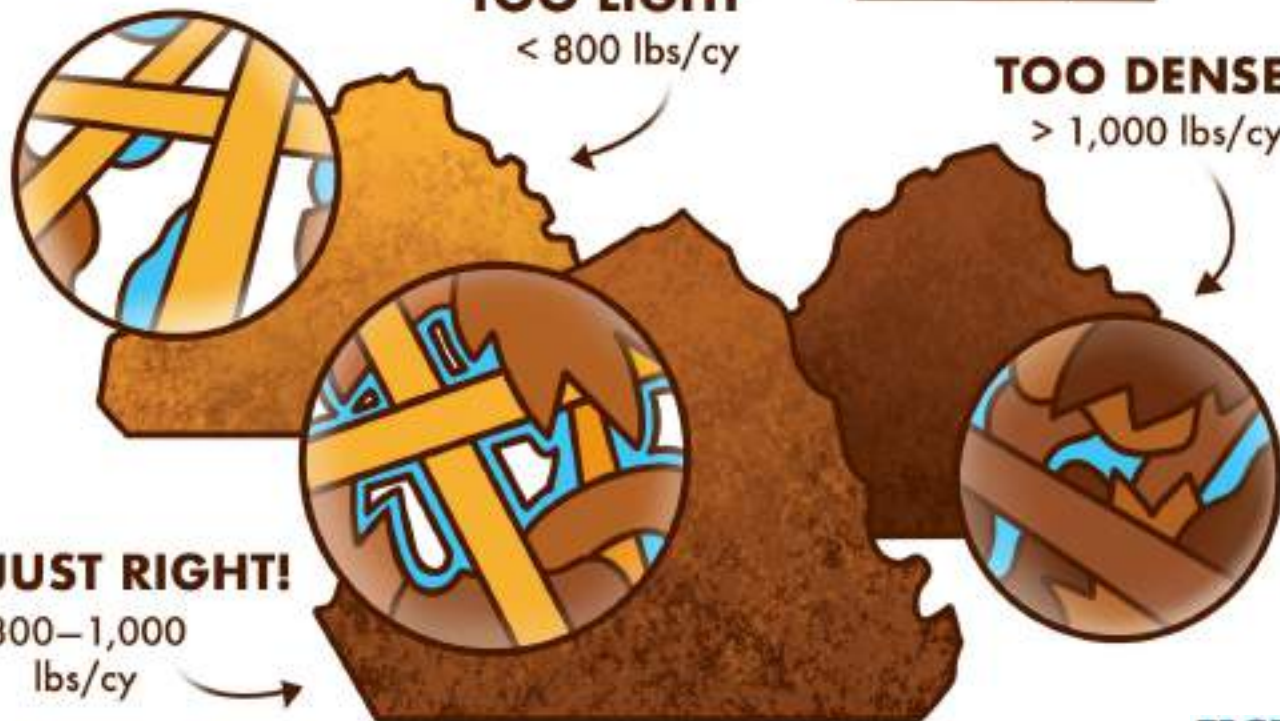
Pay Attention to Density!

● Water
 ○ Air
 lbs/cy = pounds per cubic yard

TOO LIGHT
 $< 800 \text{ lbs/cy}$

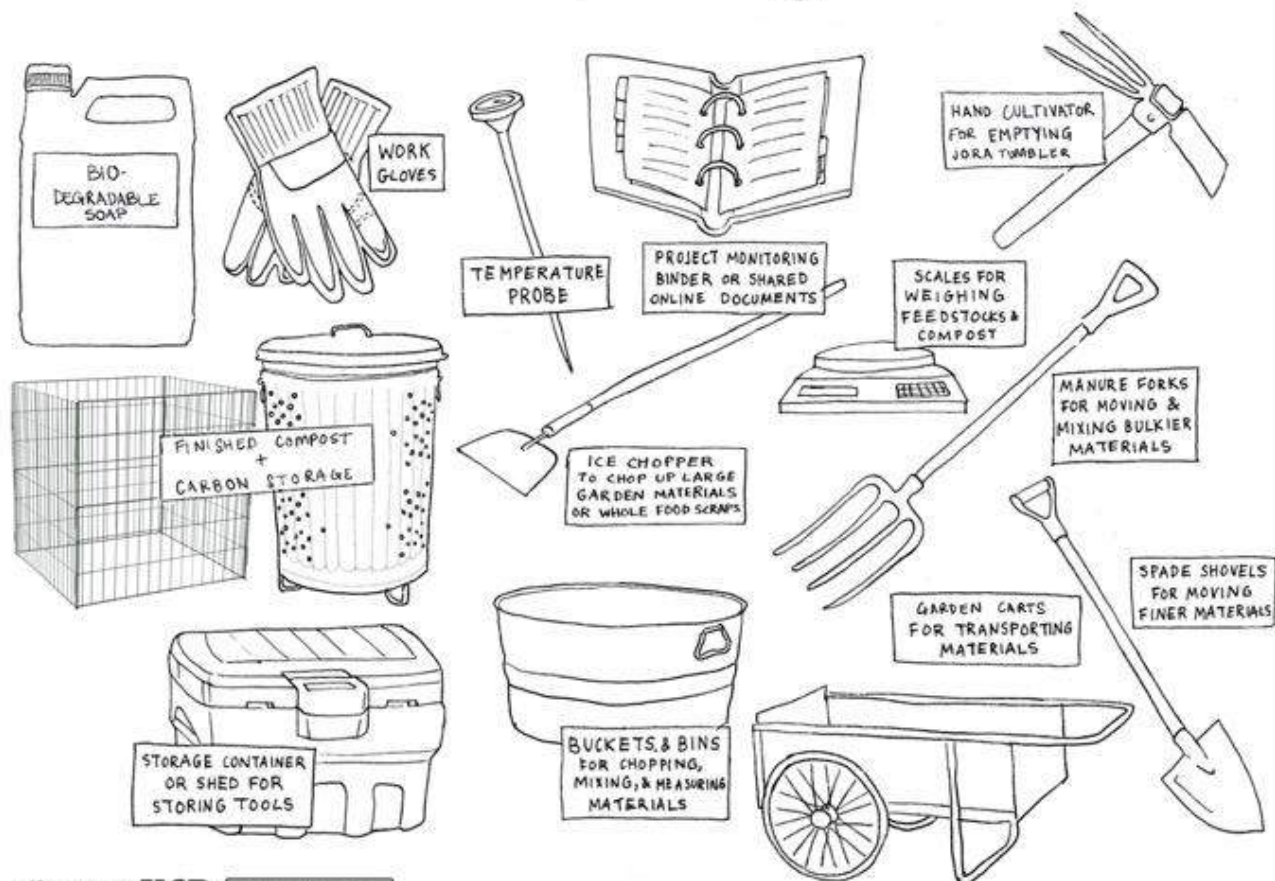
TOO DENSE
 $> 1,000 \text{ lbs/cy}$

JUST RIGHT!
 $800 - 1,000 \text{ lbs/cy}$



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Hot Composting Tools



Building a Compost Pile

Jeffrey Neal of Loop Closing at the Howard University Compost Cooperative, part of the DC Department of Parks and Recreation's Community Compost Network.

Source: Loop Closing

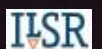




The ECUF Compost Project at East Capitol Urban Farm in Washington, D.C. Source: ILSR



Rust Belt Riders in Cleveland. Source: Jacob Koestler

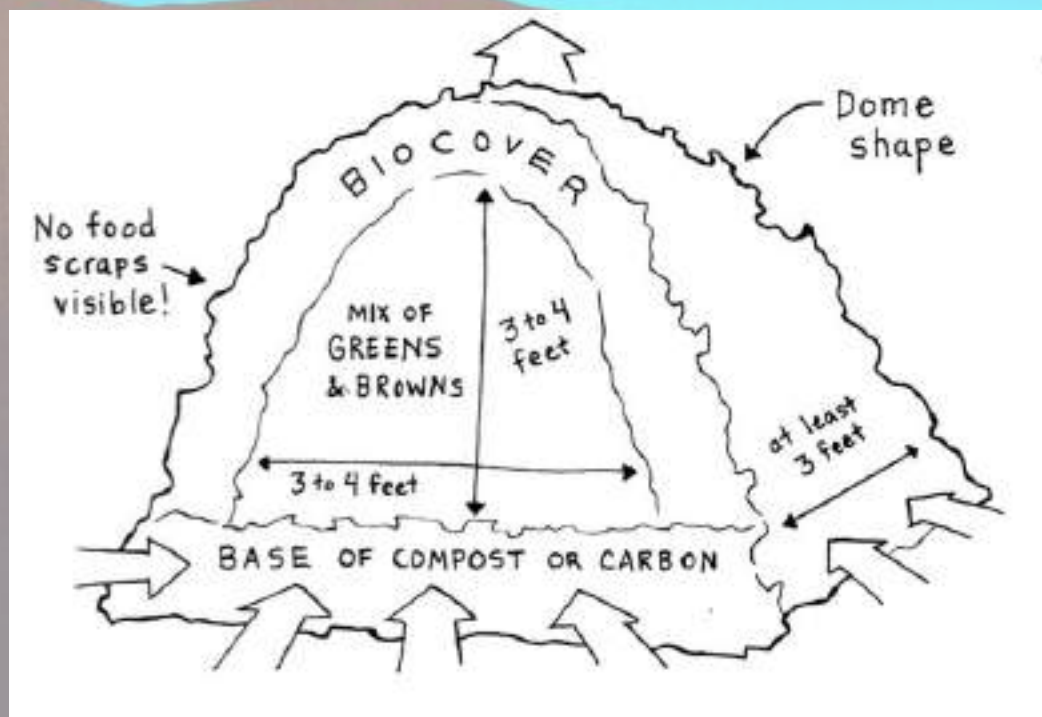




Hudson Soil Company's on-farm composting operation in the Hudson Valley. Source: Community Compost Company



Building a Compost Pile



A cross-section of a well-built composting pile.





ECO City Farms in Edmonston, Maryland.
Source: Institute for Local Self-Reliance.



The Red Hook Community Farm Compost Project
Source: David Buckel



The NYC Compost Project hosted by BIG Reuse
located under the Queensboro Bridge
Source: ILSR



Ideal Composting Conditions

Variable	Ideal Range
C:N Ratio	25:1 - 30:1
Moisture Content	50 - 60%
Particle Size	1/8 - 2 inches
Bulk Density	800 - 1,000 pounds per cubic yard
Pile Size	> 1 cubic yard

Adapted from the US Composting Council's Compost Facility Operator Training (2015)



Method 1: Lasagna Layering

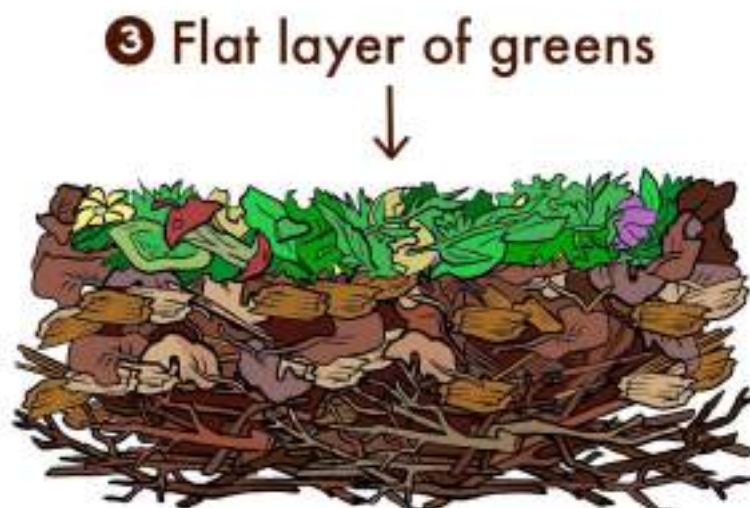
1 Layer of twigs



Method 1: Lasagna Layering



Method 1: Lasagna Layering



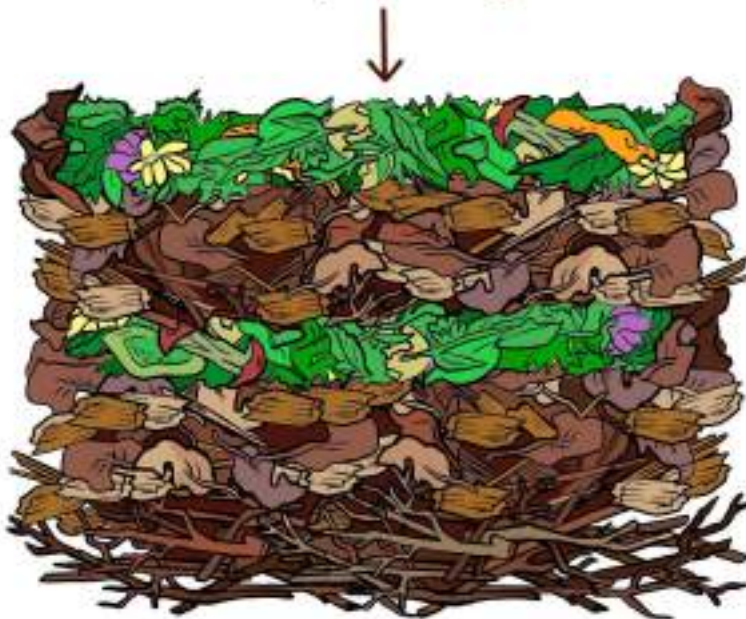
Method 1: Lasagna Layering

④ Flat layer of browns
(2-3 times thicker)



Method 1: Lasagna Layering

⑤ Flat layer of greens



Method 1: Lasagna Layering

6 Biocover



Method 1: Lasagna Layering

6 Biocover

5 Flat layer of greens

4 Flat layer of browns (2-3 times thicker)

3 Flat layer of greens

2 4-6 inches of browns

1 Layer of twigs



Method 2: Mix Greens & Browns First, Then Build

Carbon “Donut” on Paved Surface



BK ROT, NYC



Baltimore Compost Collective



37

Method 2: Mix Greens & Browns First, Then Build

Carbon “Donut” in Mixing Bin for Unpaved Surface



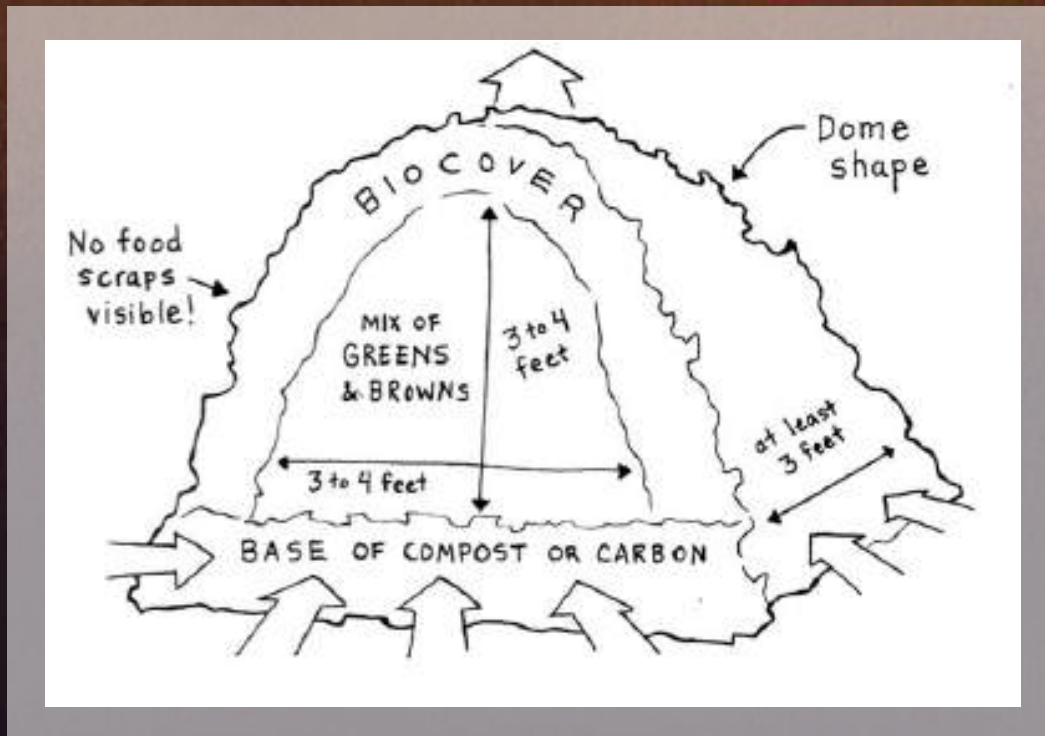
ECUF Compost Project at East Capitol Urban Farm, DC



NSR Master Composter course (2015) at Real Food Farm, Baltimore

38

Method 2: Mix Greens & Browns First, Then Build



Method 3: Fill Bin with Browns, Add Food Scraps and Mix

- 1 Layer of twigs
- 2 Fill in with browns at right moisture level



Method 3: Fill Bin with Browns, Add Food Scraps and Mix

③ Make a
small hole
in nest of
browns



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Method 3: Fill Bin with Browns, Add Food Scraps and Mix

④ Add food scraps into hole



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Method 3: Fill Bin with Browns, Add Food Scraps and Mix

⑤ Cover food scraps with browns



Method 3: Fill Bin with Browns, Add Food Scraps and Mix

- ⑤ Cover food scraps with browns
- ④ Add food scraps into hole
- ③ Make a small hole in nest of browns
- ② Fill in with browns at right moisture level
- ① Layer of twigs



Method 3: Tumblers

1

Add browns
at the right
moisture
level



Method 3: Tumblers

2

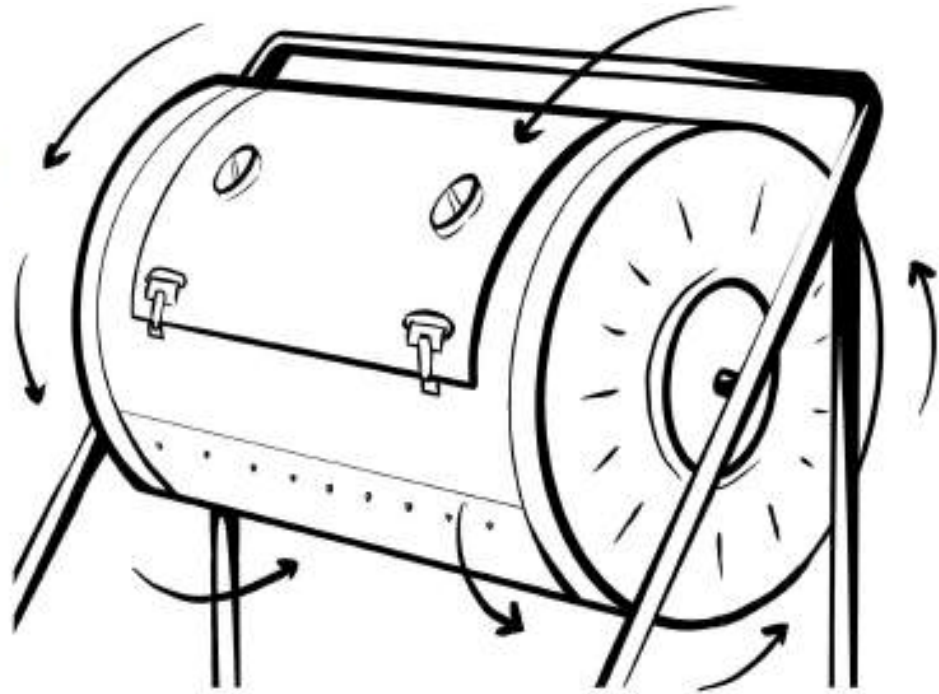
Add greens



Method 3: Tumblers

3

Tumble!



Method 3: Tumblers

4

Cover with
a layer of
browns



In Closing

- ✓ **Yes/No Materials**
- ✓ **Mix of Browns**
- ✓ **Moisture Content & Bulk Density**
- ✓ **Preparing Materials**
- ✓ **Methods for Building**



Compostable art by Jessica Hamm, Rochester, NY



Managing the Composting Process

- ✓ **During Active Composting:**
 - Monitoring, Turning, Keeping Records, Troubleshooting
- ✓ **Finishing the Process:**
 - Curing, Screening, Testing, Using Compost

